Lab 4 Prelab Assignment: Gain-Lead-Lag Compensation

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1 Introduction

In this prelab, we analyze the properties of gain-lead-lag compensators and their implementation in both continuous and discrete time. The goal is to understand how lead and lag components affect phase, gain, and system performance, and to derive implementable digital control laws.

The compensator is defined as:

$$G_c(s) = K\frac{p_d}{z_d}\frac{s+z_d}{s+p_d} \cdot \frac{s+z_g}{s+p_g}$$

where:

$$0 < z_d < p_d \quad \text{(lead)}, \qquad 0 < p_g < z_g \quad \text{(lag)}$$

2 Task 1: Phase Characteristics of Lead and Lag Compensators

2.1 Lead Compensator

The lead compensator is:

$$G_d(s) = \frac{p_d}{z_d}\frac{s+z_d}{s+p_d}$$

The phase is:

$$\phi_d(\omega) = \tan^{-1}\left(\frac{\omega}{z_d}\right) - \tan^{-1}\left(\frac{\omega}{p_d}\right)$$

Maximum phase occurs when:

$$\frac{d\phi_d}{d\omega} = 0 \Rightarrow \omega = \sqrt{z_d p_d}$$

Define:

$$\alpha_d = \frac{z_d}{p_d}$$

Maximum phase lead:

$$\boxed{\phi_{\max} = \sin^{-1}\left(\frac{1-\alpha_d}{1+\alpha_d}\right)}$$

2.2 Lag Compensator

The lag compensator is:

$$G_g(s) = \frac{s+z_g}{s+p_g}$$

Its phase:

$$\phi_g(\omega) = \tan^{-1}\left(\frac{\omega}{z_g}\right) - \tan^{-1}\left(\frac{\omega}{p_g}\right)$$

Minimum phase occurs at:

$$\boxed{\omega = \sqrt{z_g p_g}}$$

Define:

$$\alpha_g = \frac{p_g}{z_g}$$

Maximum negative phase:

$$\boxed{\phi_{\min} = -\sin^{-1}\left(\frac{1-\alpha_g}{1+\alpha_g}\right)}$$

3 Task 2: Effect of Parameter α_d and Double Lead Compensation

3.1 Effect of α_d

$$\alpha_d = \frac{z_d}{p_d}$$

As $\alpha_d \downarrow$:

- Pole and zero separation increases
- Phase boost increases:

$$\phi_{\max} = \sin^{-1}\left(\frac{1-\alpha_d}{1+\alpha_d}\right)$$

- High-frequency gain increases:

$$|G_d(j\omega)| \to \frac{p_d}{z_d} = \frac{1}{\alpha_d}$$

3.2 Double Lead Compensator

$$G_{d2}(s) = \left[\frac{p_d}{z_d}\frac{s+z_d}{s+p_d}\right]^2$$

Maximum phase:

$$\boxed{\phi_{\max,2} = 2\sin^{-1}\left(\frac{1-\alpha_d}{1+\alpha_d}\right)}$$

Frequency remains:

$$\boxed{\omega = \sqrt{z_d p_d}}$$

4 Task 3: Bilinear Discretization

Using Tustin transformation:

$$s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$$

Let:

$$A = \frac{2}{T}$$

4.1 Lead Compensator

$$G_d(z) = \frac{b_{0d} + b_{1d}z^{-1}}{1 + a_{1d}z^{-1}}$$

$$a_{1d} = \frac{p_d - A}{p_d + A}$$

$$b_{0d} = \frac{p_d}{z_d}\frac{A+z_d}{A+p_d} \quad b_{1d} = \frac{p_d}{z_d}\frac{z_d - A}{A+p_d}$$

Recursive form:

$$\boxed{x[k] = -a_{1d}x[k-1] + b_{0d}e[k] + b_{1d}e[k-1]}$$

4.2 Lag Compensator

$$G_g(z) = \frac{b_{0g} + b_{1g}z^{-1}}{1 + a_{1g}z^{-1}}$$

$$a_{1g} = \frac{p_g - A}{p_g + A}$$

$$b_{0g} = \frac{A+z_g}{A+p_g} \quad b_{1g} = \frac{z_g - A}{A+p_g}$$

Recursive form:

$$\boxed{v[k] = -a_{1g}v[k-1] + b_{0g}x[k] + b_{1g}x[k-1]}$$

Final output:

$$\boxed{u[k] = Kv[k]}$$

4.3 Stability Condition

The bilinear transform preserves stability:

$$\boxed{T > 0}$$

Practical choice:

$$T \ll \frac{1}{\max(p_d, p_g, z_d, z_g)}$$

5 Part 4: Analysis Using Identified DC Motor Model

5.1 System Model

From the previous DC motor parameter identification experiment, the open-loop position transfer function is given by:

$$G_p(s) = \frac{319.31}{s(s + 37.04)} \tag{1}$$

This model represents a Type 1 system with:

- One pole at the origin (integrator)
- One real pole at -37.04 rad/s

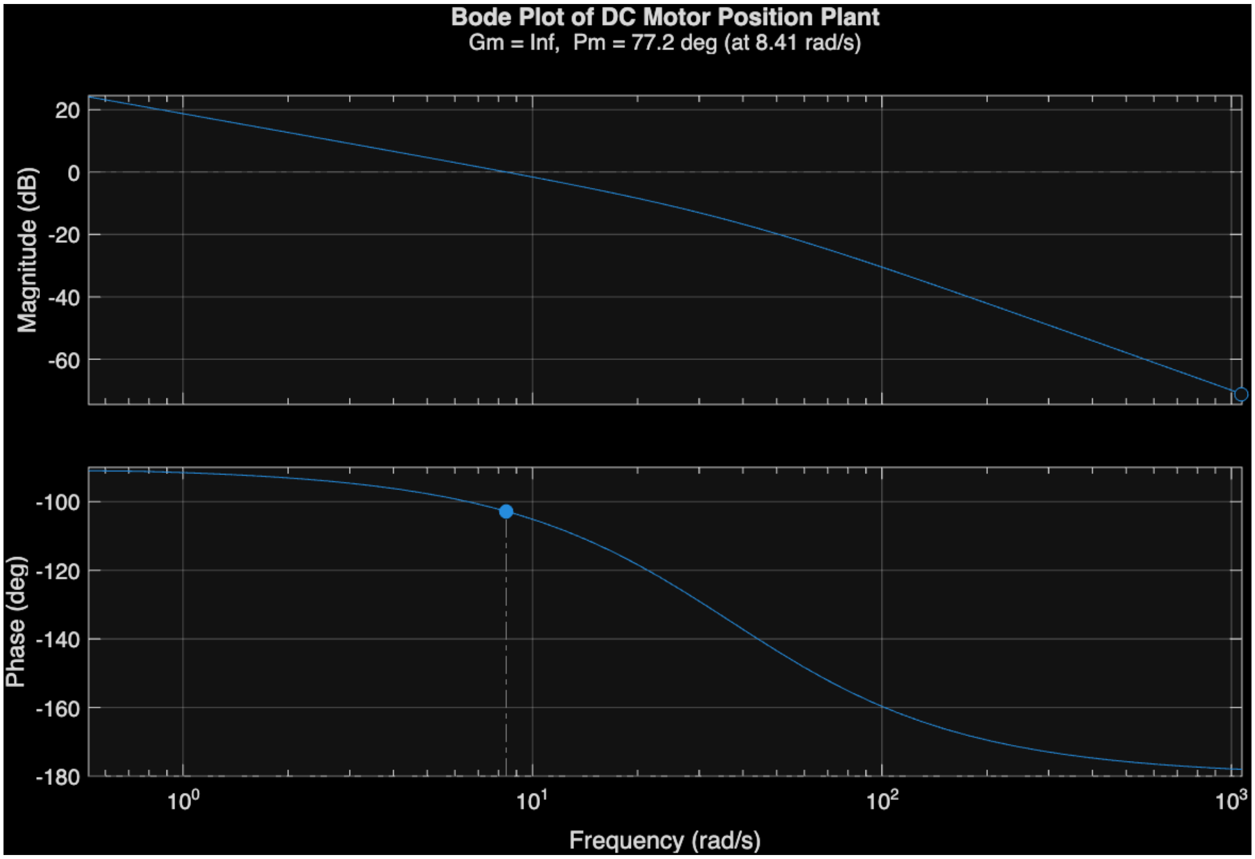
The presence of the integrator ensures zero steady-state error for step inputs, while the additional pole introduces phase lag and limits system bandwidth.

5.2 Frequency Response Analysis

The Bode plot of the uncompensated system was generated using MATLAB. The system exhibits the following characteristics:

- Initial phase near -90° due to the integrator
- Phase asymptotically approaching -180°
- Magnitude slope transitions from -20 dB/decade to -40 dB/decade

The Gain Margin (GM): Inf, Phase Margin (PM): 77.2124 deg, Gain crossover freq (Wcg): Inf rad/s and Phase crossover freq (Wcp): 8.4069 rad/s were obtained using the `margin()` function.



5.3 Closed-Loop Step Response

The unity feedback closed-loop transfer function is:

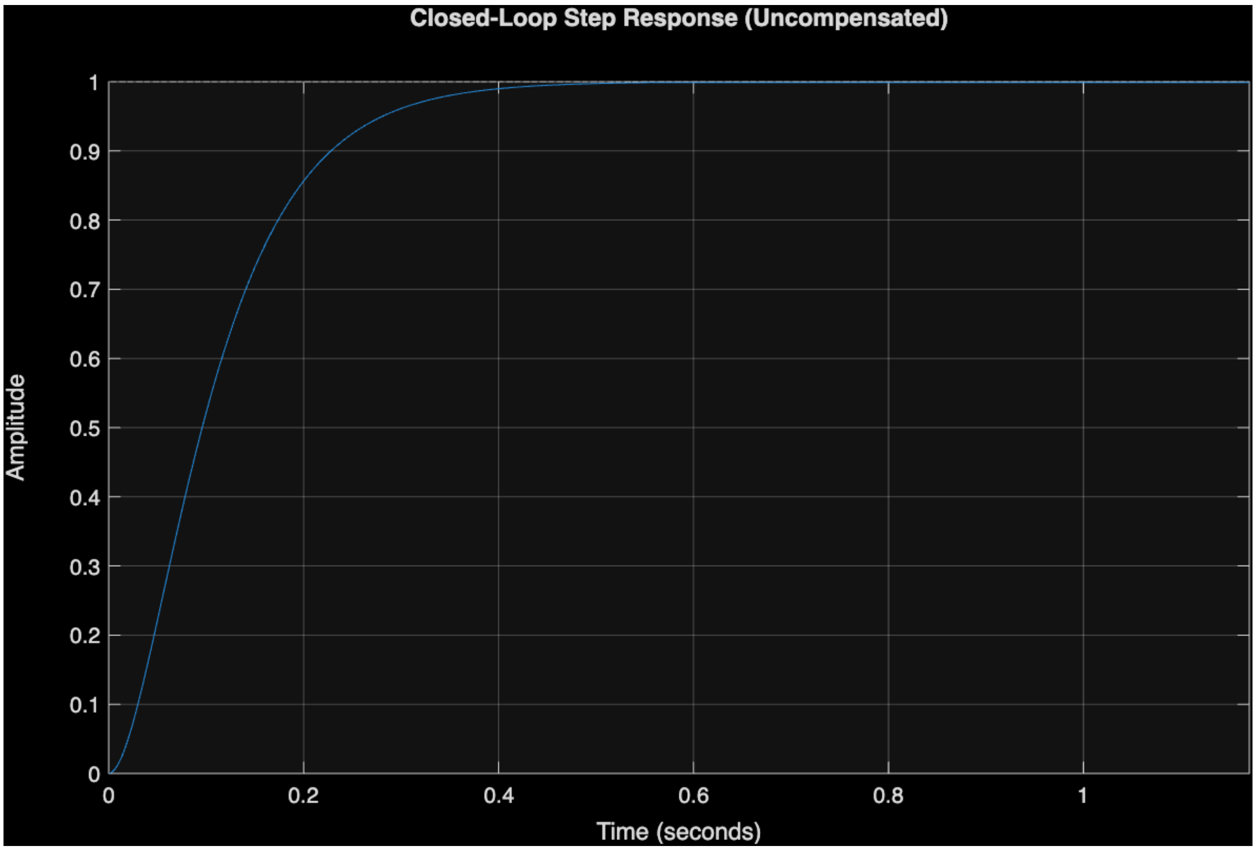
$$T(s) = \frac{G_p(s)}{1 + G_p(s)} \tag{2}$$

The step response shows:

- Stable but relatively slow dynamics
- No steady-state error due to system type
- Limited damping depending on gain selection

- RiseTime: 0.1981
- TransientTime: 0.3493
- SettlingTime: 0.3493
- SettlingMin: 0.9017
- SettlingMax: 0.9990

- Overshoot: 0
- Undershoot: 0
- Peak: 0.9990
- PeakTime: 0.5700



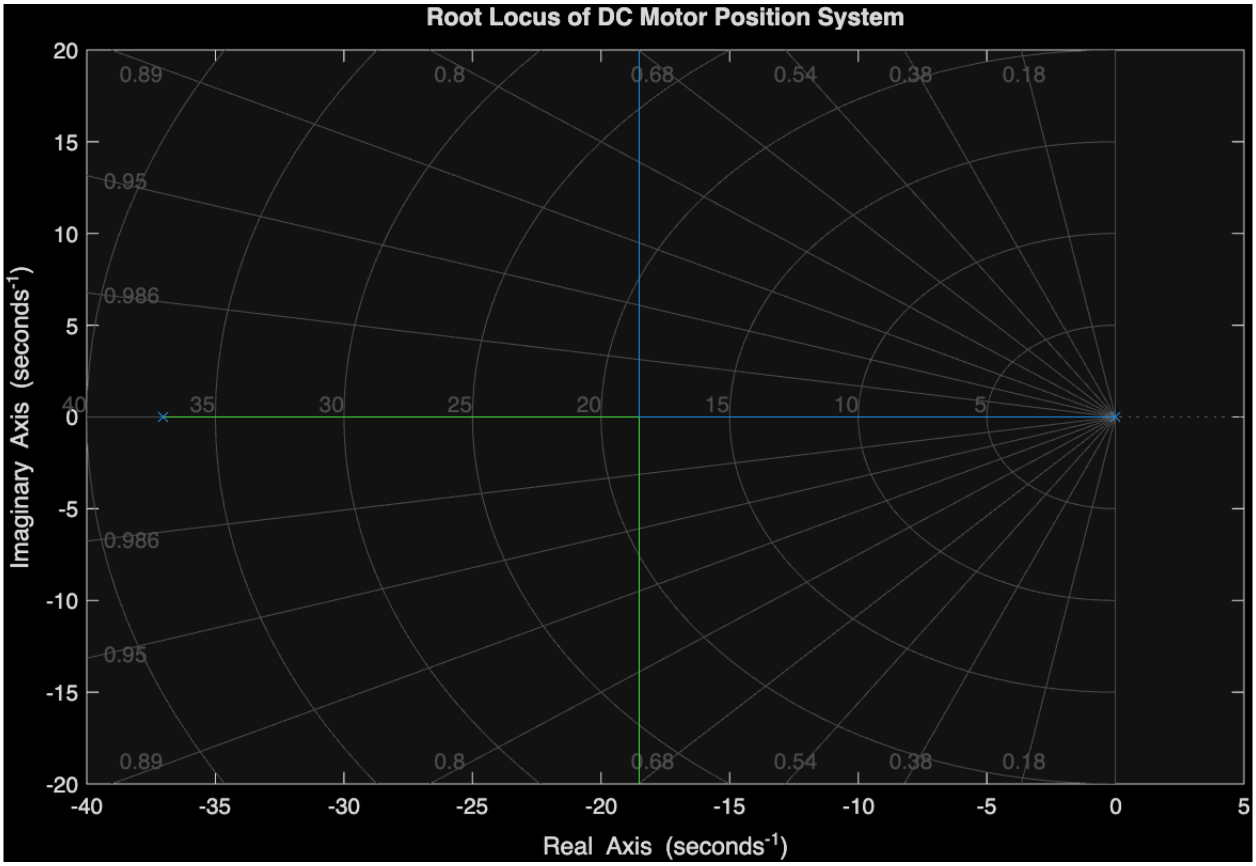
5.4 Root Locus Analysis

The root locus of the system shows the movement of poles as gain K increases.

- One pole originates at $s = 0$
- One pole originates at $s = -37.04$
- As gain increases, poles move toward each other and enter the complex plane

This behavior illustrates the trade-off between:

- Faster response (higher gain)
- Increased overshoot and reduced damping



5.5 Compensator Design Considerations

To improve system performance, a lead-lag compensator may be introduced:

$$G_c(s) = K_c \cdot \frac{s + z_d}{s + p_d} \cdot \frac{s + z_g}{s + p_g} \tag{3}$$

where:

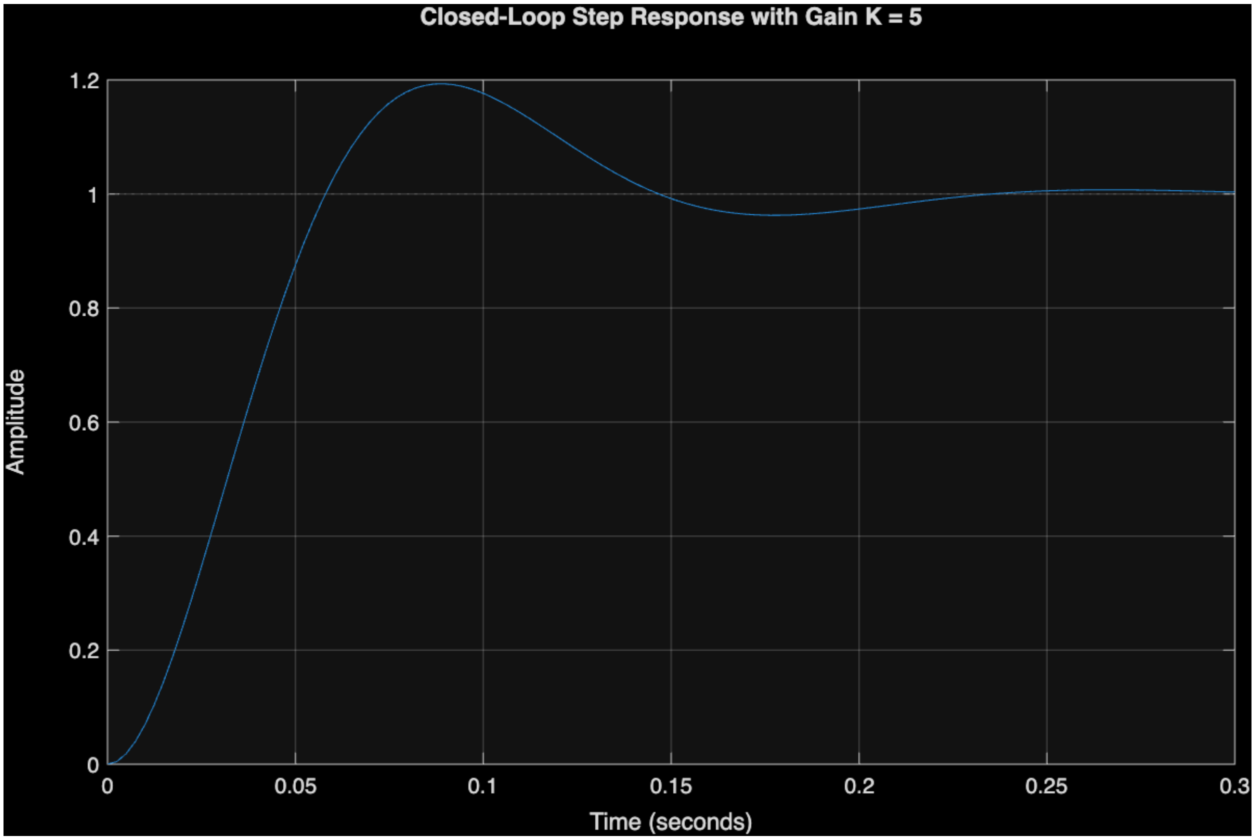
- Lead compensator ($p_d > z_d$) improves phase margin and transient response
- Lag compensator ($z_g > p_g$) improves steady-state accuracy

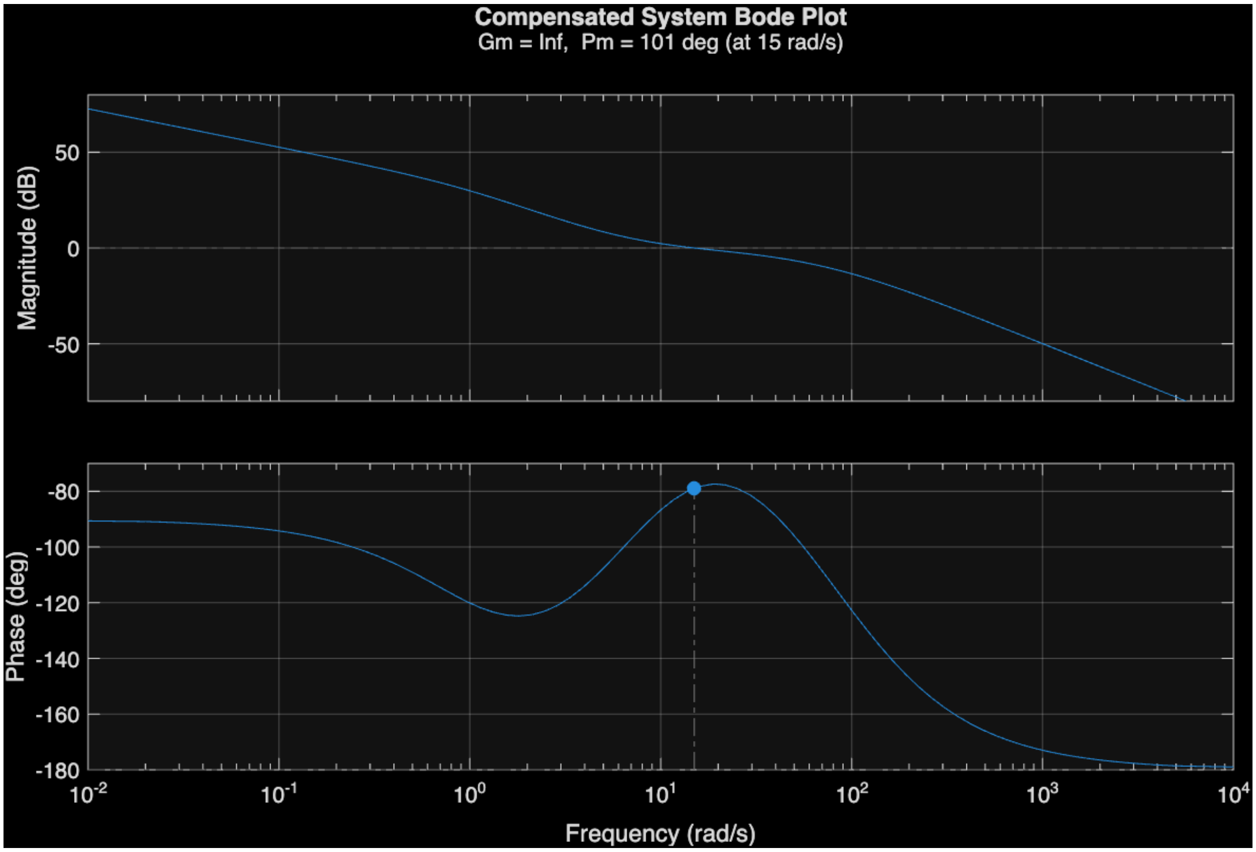
The compensated open-loop transfer function becomes:

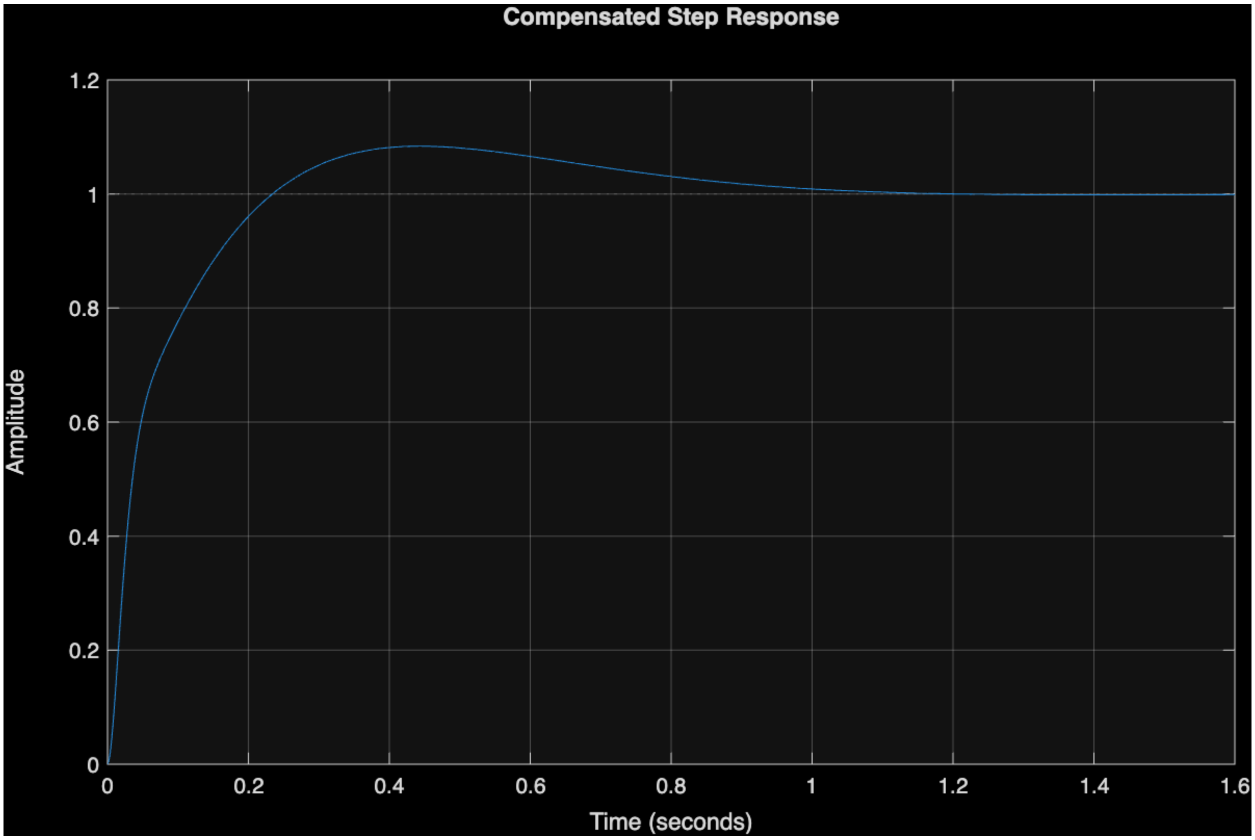
$$L(s) = G_c(s)G_p(s) \tag{4}$$

This allows tuning of:

- Phase margin
- Bandwidth
- Settling time and overshoot


Closed-Loop Step Response with Gain K = 5
Amplitude
Time (seconds)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.05
0.1
0.15
0.2
0.25
0.3


Compensated System Bode Plot
Gm = Inf, Pm = 101 deg (at 15 rad/s)
Magnitude (dB)
Phase (deg)
Frequency (rad/s)



5.6 Summary

The identified DC motor model provides a realistic plant for control design. The uncompensated system exhibits:

- Type 1 behavior (zero steady-state error for step inputs)
- Limited phase margin
- Trade-off between stability and speed

Compensation techniques such as lead or lead-lag control are necessary to achieve improved transient performance while maintaining stability.